Section 1: results of run 1

Stochastic gradient-based optimization is of core practical importance in many fields of science and engineering. Many problems in these fields can be cast as the optimization of some scalar objective function that requires maximization or minimization with respect to its parameters.

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Section 2: results of run 2

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Section 3: results of run 3

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Section 4: results of run 4

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Section 5: results of run 5

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Section 6: results of run 6

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Section 7: results of run 7

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We call the method Adam. The hyper-parameters have intuitive interpretations and typically require little tuning. Some connections to related algorithms are discussed below.

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Section 8: results of run 8

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Section 9: results of run 9

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Section 10: results of run 10

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Section 11: results of run 11

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The method is *remarkably* robust to noisy gradients.

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Section 12: results of run 12

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Section 13: results of run 13

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Section 14: results of run 14

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Section 15: results of run 15

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